

II PREPARATORY EXAM – DECEMBER 2025

Class: X

SUBJECT: BIOLOGY

Marks: 80

NAME: _____ Class & Sec: _____ Roll No. _____

SECTION – A

Attempt all the questions from this section

Question 1

A) Choose the correct answers to the questions from the given options:[15]

(i) Which of the following process occurs only in meiosis and **NOT** in mitosis?

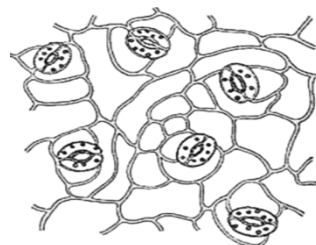
- (a) Chromosome duplication
- (b) Cytokinesis
- ☒ (c) Crossing over
- (d) Chromosome alignment

Answer: _____

(ii) The diagram below shows a minute opening in the leaf.

The function is it _____.

- (a) Helps in bleeding
- (b) Helps in guttation
- (c) Helps in imbibition
- ☒ (d) Helps in transpiration



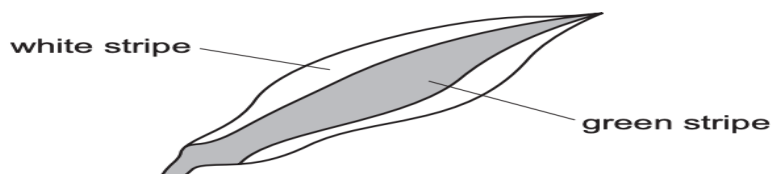
Answer: _____

(iii) Which statement about the sex chromosomes is correct?

- ☒ (a) Men and women can inherit characteristics controlled by genes on the X chromosomes.
- (b) Men and women can inherit characteristics controlled by genes on the Y chromosomes.
- (c) Only men can inherit characteristics controlled by genes on the X chromosomes.
- (d) Only women can inherit characteristics controlled by genes on the Y chromosomes.

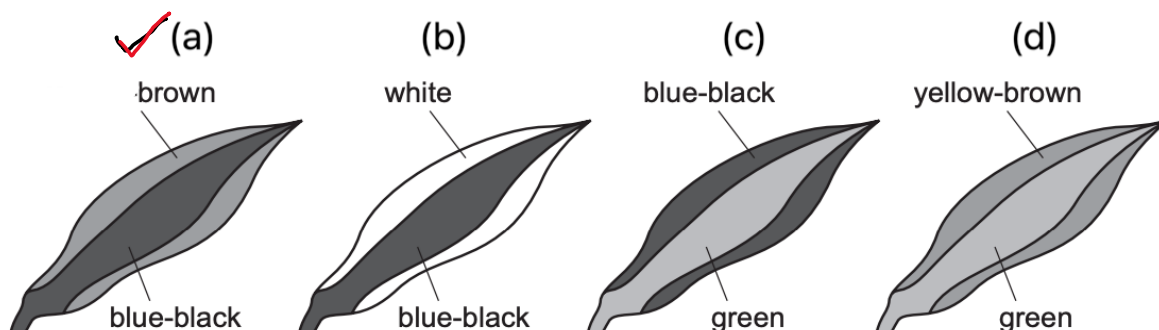
Answer: _____

- (iv) A plant with striped(variegated) leaves was kept in bright sunlight for six hours.



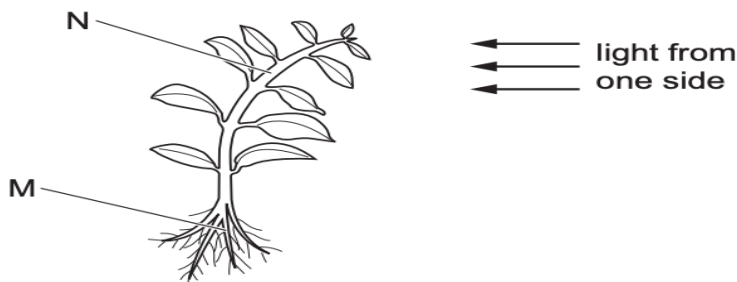
A leaf was taken from the plant and the chlorophyll was removed. The leaf was then tested for starch using iodine solution.

Which diagram below shows the result of the test?



Answer: _____

- (v) Tropic responses can be positive or negative.
When a plant grows towards the stimulus, the response is positive (+).
When a plant grows away from the stimulus, the response is negative (-).
The diagram shows a plant exhibiting tropic responses.



Which row states the tropic responses occurring at position M and at position N?

	Position M		Position N	
	Gravitropism	Phototropism	Gravitropism	Phototropism
(a)	+	+	-	-
(b)	+	-	-	+
(c)	-	+	+	-
(d)	-	-	+	+

Answer: _____

- (vi) The number of chromosomes in the somatic cells of a human body before mitosis and after mitosis
- (a) 23, 23 (c) 46, 23
- (b) 23, 46 ☒ (d) 46, 46

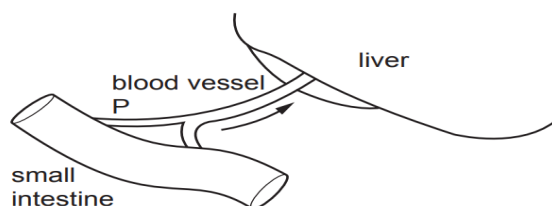
Answer: _____

- (vii) Assertion (A): Fertilization in human occurs in the uterus.
Reason (R): The uterus provides a suitable environment for the development of a fertilized egg.

- (a) (A) is True but (R) is False
- ☒ (b) (A) is false and (R) is true
- (c) Both (A) are (R) are true and (R) is the correct explanation of (A)
- (d) Both (A) are (R) are True and (R) is not the correct explanation of (A)

Answer: _____

- (viii) The diagram below illustrates blood vessel P which carries digested food from the small intestine to the liver.



Which row describes the level of glucose in blood vessel P and the level of glycogen in the liver, shortly after a meal containing carbohydrates?

	Glucose in blood vessel P	Glycogen in Liver
(a)	high	decreasing
☒ (b)	high	increasing
(c)	low	decreasing
(d)	low	increasing

Answer: _____

- (ix) Assertion (A): Guttation is the process by which plants release excess water in the form of droplets from the tips and margins of their leaves.
Reason (R): Guttation occurs when the rate of transpiration is lower than the rate of water absorption by the roots.

- (a) (A) is True but (R) is False
- (b) (A) is false and (R) is true
- ☒ (c) Both (A) and (R) are True and (R) is the correct explanation of (A)
- (d) Both (A) and (R) are False and (R) is not the correct explanation of (A)

Answer: _____

(x) Systemic circulation is

- (a) Left ventricle $\xrightarrow[\text{blood}]{\text{Deoxygenated}}$ Tissues $\xrightarrow[\text{blood}]{\text{Oxygenated}}$ Right ventricle
- (b) Right ventricle $\xrightarrow[\text{blood}]{\text{Oxygenated}}$ Tissues $\xrightarrow[\text{blood}]{\text{Deoxygenated}}$ Right auricle
- (c) Left ventricle $\xrightarrow[\text{blood}]{\text{Deoxygenated}}$ Tissues $\xrightarrow[\text{blood}]{\text{Oxygenated}}$ Right auricle
- ☒ (d) Left ventricle $\xrightarrow[\text{blood}]{\text{Oxygenated}}$ Tissues $\xrightarrow[\text{blood}]{\text{Deoxygenated}}$ Right auricle

Answer: _____

(xi) When a woman reaches the age of 50, eggs are released less often from her ovaries. To try and correct this, her body increases the production which one of the hormone given below?

- (a) Adrenaline
- ☒ (b) Follicle stimulating hormone
- (c) Progesterone
- (d) Testosterone

Answer: _____

(xii) Which of the following is responsible for causing ozone hole?

- | | | | |
|-------------|--------------------|--|---------------------|
| (a) Methane | (b) Carbon dioxide | ☒ (c) Aerosols | (d) Sulphur dioxide |
|-------------|--------------------|--|---------------------|

Answer: _____

(xiii) Anuj participated in a marathon race and felt dehydrated by the end of the race. The human body has mechanisms to conserve water and prevent dehydration. Which part of the nephron is primarily responsible for the reabsorption of water to prevent dehydration?

1. Glomerulus
2. Proximal Convuluted Tubule

3. Loop of Henle
4. Distal Convolute Tubule

- (a) Only 3 (c) Only 1, 2 and 3
 (b) Only 2 and 3 ☒ (d) Only 2, 3 and 4

Answer: _____

(xiv) Simran tried to match the ancestors with their height.

NAME	HEIGHT
P - Neanderthal	160 cm
Q - Australopithecus	120 cm
R - Homo sapiens	190 cm
S - Homo habilis	150 cm

Identify the correct sequence

- (a) S, Q, R, P
 (b) P, S, R, Q
 (c) ☒ Q, S, P, R
 (d) R, P, S, Q

Answer: _____

(xv) For the stomata of a leaf to open, the guard cells accumulate more potassium ions than the surrounding cells.

Which row describes what happens?

	Movement of K ⁺ ions	Movement of water in relation to guard cells	Final state of the guard cell
☒ (a)	Active transport	In	Turgid
(b)	Active transport	Out	Turgid
(c)	Diffusion	In	Plasmolysed
(d)	Diffusion	Out	No change

Answer: _____

Question 2

(i) Give the biological/technical terms for the following: [5]

- (a) The pollutants that can be broken down to simple and harmless products- **Biodegradable wastes**
- (b) A plant with sunken stomata- **Nerium**
- (c) The wax-like layer on the epidermis of the leaves which reduces transpiration- **Cuticle**
- (d) The process of transformation of several glucose molecules into one molecule of starch- **Polymerization**
- (e) The glands located at the upward sideward portion of the orbit of the eye- **Lacrimal glands**

- (ii) Choose the odd term out from each of the following set of terms.
Mention the category to which the remaining three belong: [5]

(a) Detergents, X-rays, Sewage, Oil spills

A. X-rays. Water pollutants

(b) Bud dormancy, Senescence, Fruit ripening, Gaseous hormone

A. Bud dormancy. Functions and features of ethylene

(c) Sulphur dioxide, Carbon monoxide, Oxygen, Nitrogen dioxide

A. Oxygen. Gaseous pollutants

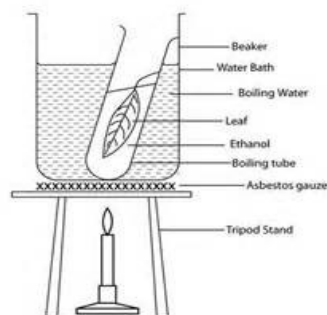
(d) Temperature, Humidity, Wind velocity, Fewer stomata

A. Fewer stomata. External factors affecting transpiration

(e) Renal pelvis, Renal cortex, Renal artery, Renal medulla

A. Renal artery. Parts of the Kidney

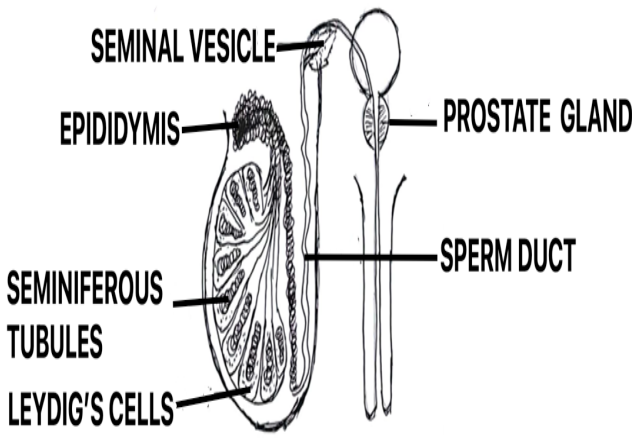
- (iii) The diagram indicates a step involved in Iodine test.
Read the information below the diagram and fill in the blanks: [5]



Plants make glucose in leaves during photosynthesis but leaves cannot be tested for its presence as glucose is quickly used, stored as starch, converted into sucrose, fats, proteins and other substances. Testing the leaf for starch is a reliable indicator.

To test a leaf for starch, the leaf is boiled in water to (a) kill the cells. It is then boiled in methylated spirit to (b) remove chlorophyll. The leaf is dipped in warm water to soften it. It is placed in a petri dish and (c) iodine solution is added to it. The region of the leaf which contains starch, turns (d) blue-black and the region which does not contain starch, remains (e) brown.

- (iv) Study the diagram given below and match the structure with its functions:[5]
Eg: Epididymis – (f)

L.S.OF TESTES		FUNCTIONS
	(a)	Neutralizes the acid content in the female's vagina.
	(b)	Produce sperms.
	(c)	Carry the sperm from epididymis to urethra.
	(d)	Produce testosterone.
	(e)	Serves as a medium for transportation of sperms.
	(f)	Stores the sperms temporarily until they are mature and motile.

- A. Prostate gland – (a)
Seminiferous tubules – (b)
Sperm duct – (c)
Leydig's cell – (d)
Seminal vesicle – (e)

- (v) Seema observed the experiment conducted by her biology teacher in the school laboratory using three different solutions and three potato cubes.

Solution A: Pure distilled water

Solution B: Concentrated sugar solution

Solution C: A solution with sugar concentration similar to the potato cubes cell sap

The potatoes were placed in each solution and observed after four hours.

In Solution A, the potato cubes swelled up and became plump.

In Solution B, the potato cube shriveled and became smaller.

In Solution C, the potato cube showed no visible change.

Her teacher gave a questionnaire that explains the experiment.

Help Seema answer the questions:

[5]

- (a) Identify the tonicity of each solutions (A,B and C) with respect to the potato cell sap.

A. A- hypotonic B- hypertonic C- Isotonic

- (b) Explain why the potato cubes swelled up in Solution A.

A. As solution A is hypotonic, endosmosis takes place and the potato cubes swell up.

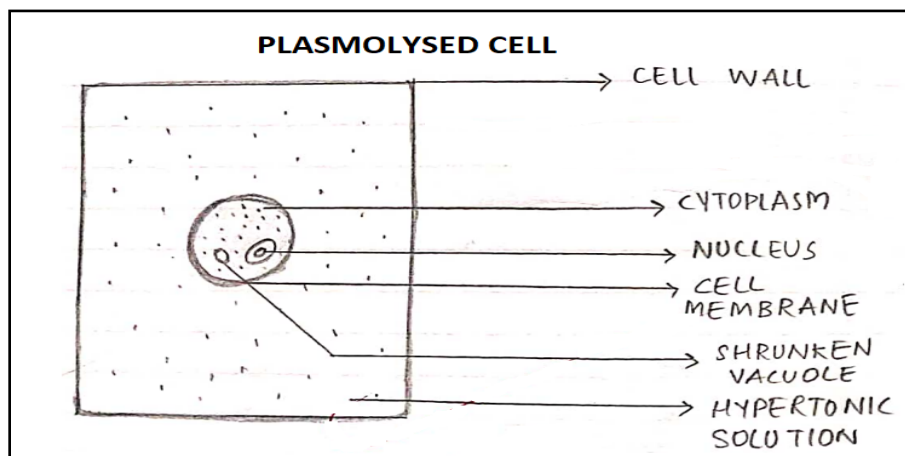
- (c) In which solution was there no net movement of water? Why?

A. Solution C as it has the sugar concentration similar to the cell sap of potato cubes.

(d) What is tonicity?

A. Tonicity is the relative concentration of the solution that determines the direction and the extent of diffusion.

(e) With the help of a neat labelled diagram, show how a potato cube cell appears in Solution B.



SECTION - B (40 marks)

(Attempt any **FOUR** questions from this Section)

Question 3

(i) Explain the term Mortality.

[1]

A. The number of deaths per thousand people per year.

(ii) State the rate of transpiration with a reason when the air surrounding the plant is:

[2]

(a) Hot and dry- More as evaporation from the surface of the leaves is faster.

(b) Humid- Less as the air is more saturated with water vapour and it is difficult for the air to accommodate water vapour.

(iii) Param was told to buy a blue cloth by his craft teacher. He couldn't differentiate between blue colour from green and hence brought a green cloth to class. [2]

(a) Which medical reason could have resulted in this?

A. Colour blindness

(b) Haemophilia is generally seen in males. Give reasons.

A. X-linked inheritance is due to the presence of a recessive gene on X chromosome. Since there is no dominant gene on Y chromosome, the recessive gene on X chromosome (X^c) is expressed and hence colour blindness is more commonly found in males. for colour blindness, which is rare.

(iv) Differentiate between the following as per the indications given in the brackets:

[2]

(a) Transpiration and Guttation [Structure]

A. Transpiration – Stomata, Lenticels, Cuticle

Guttation- Hydrathodes

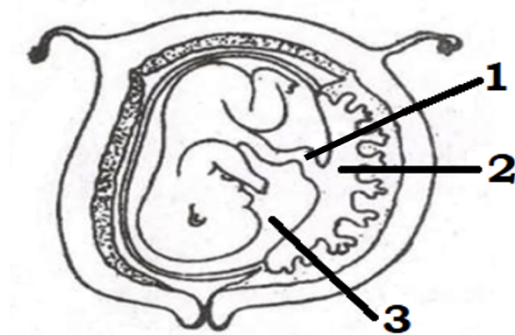
(b) Mitral valve and Aortic semilunar valve [Location]

A. Mitral valve – Between left auricle and left ventricle

Aortic semilunar valve- Between aorta and left ventricle

(v) The diagram given below represents the growth of human foetus. Study the diagram and answer the questions the follow:

[3]



(a) State the location of part 1.

A. Part 1- Umbilical cord. Between the foetus and placenta.

(b) What is the significance of part 3?

A. Part 3- Amniotic fluid

- Protects the embryo from physical damage by jerks or mechanical shocks due to fall of the mother.
- Keeps an even pressure all around the embryo.
- Allows the foetus some restricted movement.
- Prevents sticking of the foetus to the amnion.

(c) Write one function of part 2.

A. Part 2 – Placenta

- Allows diffusion of substances from mother to foetus and
Eg: Diffusion of nutrient and oxygen from mother to foetus.
Diffusion of carbon dioxide, nitrogenous wastes from foetus to the mother.
- It acts as an endocrine gland – it produces hormones called Oestrogen and Progesterone.

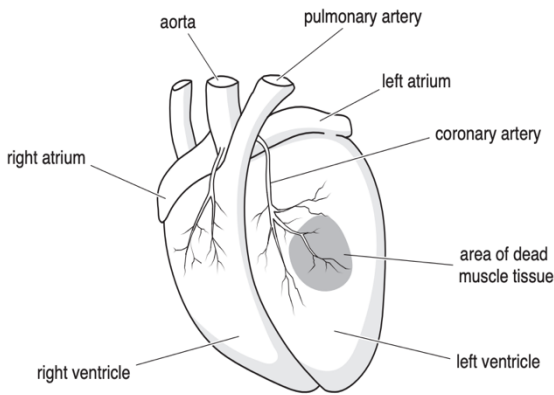
Question 4

(i) Grass growing in the shade under the stone turns yellow. Give reasons.

[1]

A. Grass growing under the shade does not receive light hence there is no formation of new chlorophyll and disintegration of the older one turns the grass yellow.

- (ii) The diagram shows an external view of the heart of a person who has coronary heart disease. Study the diagram and answer the questions given alongside: [2]



- (a) Identify the disease. Mention the reason for deadening of the heart muscle tissue.

A. Myocardial infarction. Blockage in the coronary arteries.

- (b) One factor that increases the risk of developing coronary heart disease is having a diet that contains a lot of fat. State another factor.

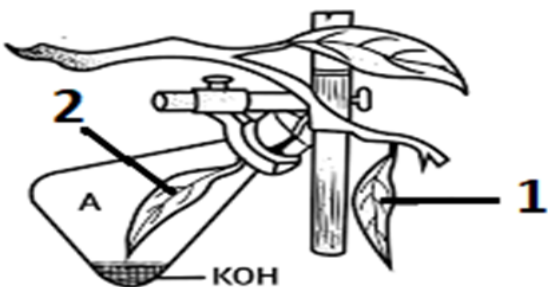
**A. Sedentary lifestyle
Stress
Obesity
Smoking
Hypertension
Diabetes
Family history (Heredity)**

- (iii) What is parthenocarpy? Give an example. [2]

**A. The development of fruits without fertilization is called parthenocarpy.
Eg: Bananas, tomatoes, apples.**

- (iv) Expand: (a) NADP – **Nicotinamide Adenine Dinucleotide Phosphate** [2]
(b) LH – **Luteinizing hormone**

- (v) The diagram given below represents an experiment to demonstrate a particular aspect of photosynthesis. The letter 'A' indicates a certain condition inside the flask. Study the diagram and answer the questions that follow: [3]



- (a) State the aim of the experiment.

A. Carbon dioxide is necessary for photosynthesis.

- (b) What is the special condition inside the flask?

A. KOH is present which absorbs carbon dioxide.

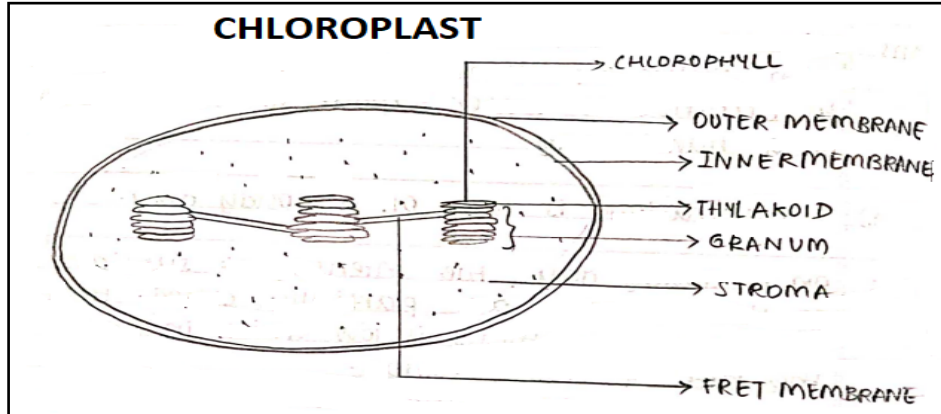
- (c) In what manner do the leaves 1 and 2 differ at the end of the starch test?

**A. Leaf 1- Turns blue-black due to the presence of starch.
Leaf 2 – Remains brown due to the absence of starch.**

Question 5

- (i) Define the term phytohormone. [1]
A. Phytohormones are chemical compounds secreted in one area of the plant body, transported throughout the plant body and have their effects at a location far away from the site of production.

- (ii) Draw a neat labelled diagram of a chloroplast. [2]



- (iii) A teacher drew the diagram of heart on the black board and told the students to copy it in their notebooks. Manish couldn't see the diagram clearly as it appeared blurred to him. [2]

- (a) Name the defect of the eye Manish is suffering from and state one reason for the same.

A. Myopia.

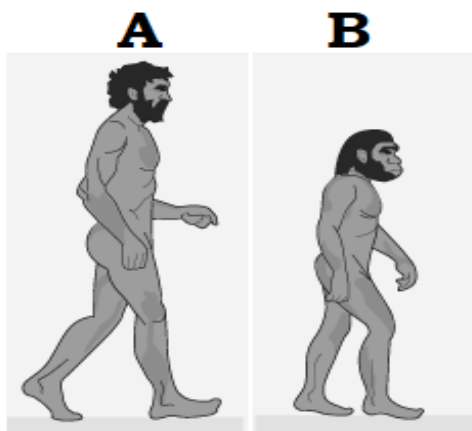
Reason:

- **Lens is too curved.**
- **Lengthening of the eye ball from front to back.**

- (b) Manish consults an ophthalmologist and is prescribed suitable lenses to correct the defect. Which type of lens do his spectacles have?

A. Concave

- (iv) Given below are two stages in the evolution of man. Study the diagram and answer the questions: [2]



(a) Identify Australopithecus and Homo sapiens.

A. A- Homo sapiens B- Australopithecus

(b) State the difference between A and B with respect to cranial capacity.

A. Cranial capacity- A- 1450 to 1600 cm³

B- 450 TO 600cm³

(v) The amazon rain forest is referred to as the 'Lungs of the Earth'. It plays an important role in climate regulation through photosynthesis.

However, this delicate ecosystem is facing threat from deforestation, climate change and human encroachment which can lead to irreversible environmental consequences. [3]



(a) How does photosynthesis contribute to the Amazon rain forests' reputation as the 'Lungs of the Earth'?

A. Oxygen is liberated to the atmosphere as a byproduct of photosynthesis and they absorb carbon dioxide from the atmosphere for photosynthesis just like the lungs.

(b) In what way is the environment affected by deforestation?

A.

- Loss of Biodiversity
- Climate change
- Soil erosion
- Disrupted water cycles
- Increased flood risk
- Global warming
- Melting of ice caps leading to rise in sea level

(c) Write a short note on Euro Bharath Vehicular Standards.

A.

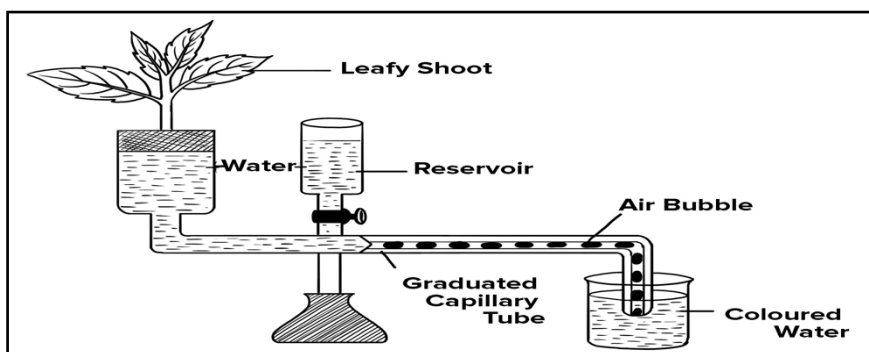
- Euro/Bharat norms aim to effectively cut down sulphur and nitrogen oxides from automobile exhausts.

- The Bharat stage (BS) emission norms are based on European emission norms which are referred to as Euro 4, Euro 5, Euro 6 etc. These norms act as a good reference point as to how much does a vehicle pollute.

Question – 6

- (i) When an ovum gets fertilized, menstrual cycle stops temporarily in a woman. Give reasons. [1]
- A. Menstruation does not occur during pregnancy because high levels of progesterone in the blood stream does not facilitate production of eggs.
- (ii) Explain the term plasmolysis. Give one application of the phenomenon in our daily life. [2]
- A. **Plasmolysis:** The contraction of the cytoplasm from the cell wall caused due to withdrawal of water when placed in a hypertonic (strong) solution.
1. Addition of salt to pickles and sugar to jam kill fungal spores and bacteria by plasmolysis.
 2. Plasmolysis is also involved in killing of weeds in lawns, orchards and agricultural fields by chemical weedicides.
 3. It helps in preventing the growth of plants in the cracks of the walls. Salting of such plants induces plasmolysis, and the unwanted plants are killed.
- (iii) Differentiate between the following as indicated in brackets: [2]
- (a) Vasectomy and Tubectomy [Gender and Part ligated]
- A. **Vasectomy-** In males, Vas deferens or Sperm duct is ligated.
Tubectomy- In females, Oviduct or Fallopian tube is ligated.
- (b) Photolysis and Photophosphorylation [Explain]
- A. **Photolysis:** The process of splitting of water molecules into hydrogen ions and oxygen in the presence of light which takes place in the grana.
Photophosphorylation: The process in which the electrons are used in converting ADP to ATP by adding one phosphate group in the presence of light which takes place in the grana.
- (iv) What is Bio medical waste? Why is it considered harmful? [2]
- A. Wastes generated from the hospitals like syringes, dirty dressings etc. which causes soil pollution.
 It contains pathogens and harmful chemicals which act as agents to spread infectious diseases.

(v) Study the given diagram and then answer the questions that follow: [3]



1. Name the apparatus and state the phenomenon demonstrated by the apparatus.

A. Ganong's potometer. Transpiration

2. What is the function of the reservoir?

A. Water in the reservoir moves to the capillary tube and brings the air bubble back to its position.

3. State one limitation of using this apparatus.

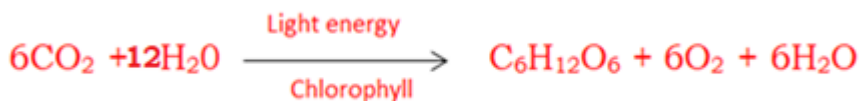
A.

- Introducing the air bubble is not very easy.
- The twig may not remain fully alive for a long time.
- Any change in the outside air temperature may affect the position of the air bubble in the capillary tube.

Question – 7

(i) Write balanced chemical equation of photosynthesis. [1]

A.



(ii) Megha observed a permanent slide showing a phase of mitotic cell division. She noticed several thread-like structures pulling apart to poles of the cell.

[2]

Diagrammatic representation



Under light microscope



(a) Identify the phase observed by Megha by giving suitable reason.

A. Anaphase

- Centromere divides
- Sister chromatids move towards the opposite poles by the shortening of the spindle fibres.

(b) How many chromosomes will each daughter cell have after the completion of the above division?

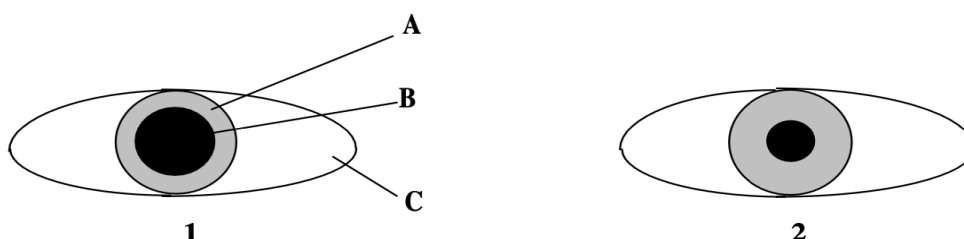
A. Four

(iii) List two main reasons for the increase in India's population. [2]

A.

- Illiteracy.
- High infant mortality rate.
- Religious and traditional beliefs.
- Desire for a male child.
- Economic reasons.

(iv) The diagrams show the front of the eye. [2]



(a) Name the parts A and B.

A. A- Iris B - Pupil

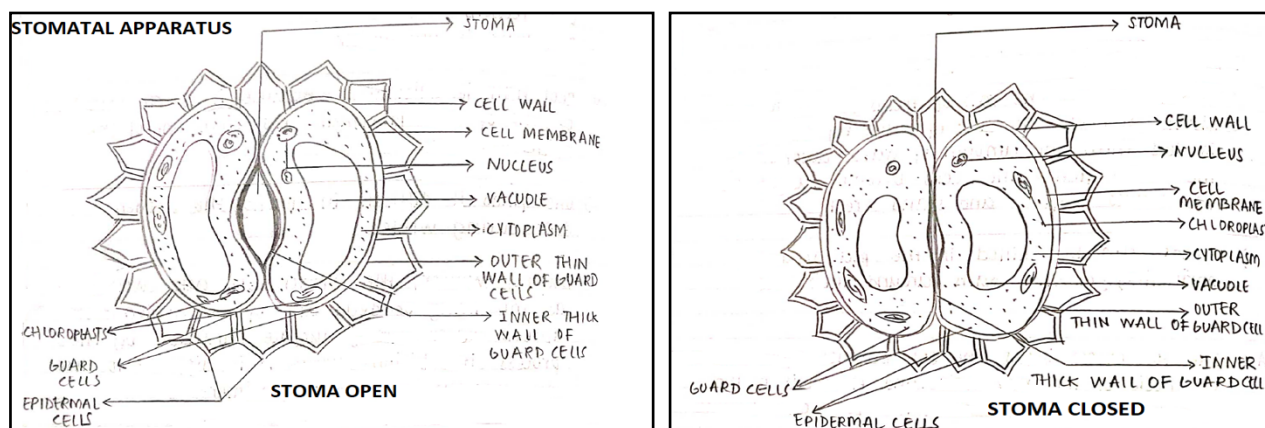
(b) What causes the eye to change from the condition shown in diagram 2 to that shown in diagram 1?

A. Dark adaptation

- Pupil dilates to allow more light to enter the eye.
- Rhodopsin is regenerated and iodopsin is degenerated (Rods are active).

(v) Draw a neat labelled diagram of the stomatal apparatus. [3]

A.



Question – 8

(i) State Mendel's Law of Dominance. [1]

A. Law of Dominance: Out of a pair of contrasting characters present together, only one is able to express itself while the other remains suppressed.

(ii) Mention two objectives of Swachh Bharath Abhiyan. [2]

A.

- To clean the streets, roads and infrastructure of the country's cities and towns.
- To eliminate open defecation through the construction of individual, cluster and community toilets. The goal is to make an ODF (open defecation free)
- Establish an accountable mechanism of monitoring latrine use.
- To achieve efficient solid and liquid waste management systems.

(iii) Bhaskar visited the dentist due to frequent toothache and the doctor explained that it was caused by an unwanted tooth appearing at the back of his jaw. [2]

(a) Name the tooth causing pain to Bhaskar.

A. Wisdom tooth

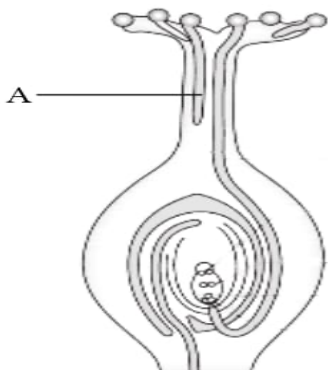
(b) What are these types of organs called? Define the same.

A. Organs that have ceased to be of any use to the possessor but still persists generation after generation in a reduced form.

OR

Vestigial organs. Vestigial organs are the remnants of features that served important functions in the organism's ancestors.

(iv) The diagram given below shows a type of tropism. Study the same and answer the questions that follow: [2]



(a) Name and define the type of tropism shown in the diagram alongside.

A. Chemotropism.

The movement of plant parts in the direction of the stimulus chemicals is called chemotropism.

(b) Give an example of a stimulant that helps in the growth of the part marked A.

A. Peptones, Sugar

(v) A pea plant which is homozygous for green pods which are inflated [GGII] is crossed with a homozygous plant for yellow pods which are constricted [ggii]. Answer the following questions: [3]

(a) Give the phenotype and genotype of F1 generation.

A. Phenotype – All pea plants have green and inflates pea pods.
Genotype - GgIi

(b) State the phenotypic ratio of F2 generation.

A. Phenotypic ratio – 9:3:3:1

(c) Write the possible combinations of the gametes that can be obtained if two F1 hybrid plants are crossed.

A. GI, Gi, gI, gi
